Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/08/2015
PI Name:	Rickus, Jenna Ph.D.		
Project Title:	Sporesat Mission: Investigating biophysical mechanisms of plant gravisensing using a lab-on-a-chip approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture (2) Bioregenerative Life Support		
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Zip Code:	47907	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	03/01/2011	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Small Satellites		
Flight Assignment:	NOTE: Extended to 9/30/2014 per NSSC information and A. Chu/NASA ARC (Ed., 1/6/2014)		
Key Personnel Changes/Previous PI:	Previous PI: Marshall Porterfield. PI change to Jenna Rickus in November 2012. Amani Salim replaced him initially until November 2012 and she remains as Co-PI. Dr. Stan Roux, a consultant and Co-I on the original proposal, was formally added to the Science Team as a Co-I.		
COI Name (Institution):	Salim, Amani (Purdue University/ Co-Principal Investigator) Roux, Stanley (University of Texas, Austin)		
Grant/Contract No.:	NNX11AF48A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A detailed understanding of the molecular and biophysical mechanisms for plant gravity sensing still eludes the scientific community. To gain deeper knowledge of the mechanism of cell gravity sensing, we will study activation of cell gravisensing and electrophysiological signaling in a single-cell model system (Ceratopteris richardii spores) using a "lab-on-a-chip" microsensor technology platform. We here propose a Small Complete Mission (SCM) based on a Mission of Opportunity-2 nanosatellite we call SporeSat, which will adapt hardware and technologies from previous and current missions including: GeneSat, PharmaSat, O/OREOS, and GraviSat. Our approach allows for real-time measurement of calcium signaling and maintenance of variable gravity treatments during the flight experiment. Our science objectives further current work that has documented molecular components of biomechanical transduction. Our technical objectives focus on refining our current lab-on-a-chip sensor system to build a new CEL-C Advanced bioCD designed for the micro-satellite platform. This project contributes to NASA's initiatives in the areas of both Astrobiology and Space Biology and will advance the science and technology capabilities for future research and development of astronaut health programs as well as those for advanced life support for exploration in the space environment, the moon, and on other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Astronaut health in long term space missions is likely to be dependent on and benefit from plants. On Earth we depend on plants for food, clean air, and waste treatment. The same will be true in space. If we as humans are to achieve the kind of space exploration and habitation that we all imagine and dream of, we are going to need plants up there with us. Yet on the International Space Station, gravity is less than one one-thousandth of the gravity here on Earth. On the Moon, it is about 17% of Earth's gravity. In addition, there are astonishing similarities in signaling within cells across species. As in plants, calcium plays important roles in human cell development, function, and dysfunction. Understanding gravity sensing in plant cells is likely to give clues to gravity sensing in other organisms including humans. In addition, the technology that we developed for these experiments establishes a platform to ask other questions regarding human cell responses on Earth and in space. We are already looking at how we can adapt the technology to study pancreatic beta cells and diabetes.
Task Progress:	Overview. SporeSat is a Small Complete Mission based on a Mission of Opportunity-2 nanosatellite designed for real time measurement of calcium signaling and maintenance of variable gravity treatments during flight. The impetus for SporeSat is to gain a deeper knowledge of the molecular and biophysical mechanisms for plant gravity sensing by studying the activation of cell gravity-sensing and electrophysiological signaling in the single-cell model system, Ceratopteris richardii, using a novel ‘lab-on-a-chip’ microsensor technology platform called a CEL-C advanced bioCD. The science and technical objectives as outlined in the original SporeSat proposal: Science Objectives: 1) Determine gravity thresholds for Ca2+ channel activation in wild-type spores; 2) Observe Ca2+ current response of gene knockout mutants. Technology Objectives: 1) Design and fabricate the CEL-C advanced bioCD; 2) Integrate bioCD with data acquisition hardware and ground-based validation experiments; 3) Develop and integrate centrifuge hardware; 4) Operate bioCD in space-relevant environment; 5) Conduct spaceflight experiments. State of SporeSat. In 2013, SporeSat was adapted into a 2 flight mission strategy with flight 1 focused on the technology objectives and flight 2 focused on the scientific objectives. In April 2014, SporeSat-1 was deployed from a SpaceX Falcon 9 launched from Cape Canaveral as part of CRS-3. SporeSat-1 was successfully commanded and science data downloaded. A summary of the current accomplishments against the original objectives after launch of SporeSat-1 is included below. Technical Objectives 1-3 – Achieved. 1) Design and fabricate the CEL-C advanced bioCD; 2) Integrate bioCD with data acquisition hardware and ground-based validation experiments; 3) Develop and integrate centrifuge hardware SporeSat is a nanosatellite (10x10x34 cm, 5.5 kg), the payload of which integrates motors, thermal control, measurement electronics, and various sensors with three bioCDs, all housed in a container with an internal pressure of 1-atm. Two bioCDs rotate; the third remains stationary as a control. A bioCD, fabricated on 2-inch fused-silica substrate, includes thirty-two 126-µm-diameter cylindrical SU8 wells (1:1 aspect ratio), each housing one fern spore. Four Pt resistance-temperature devices (RTDs) per bioCD provide closed-loop thermal control. A pair of ion-selective electrodes (ISEs) measures the ratio of Ca2+ concentrations at either “end” of each fern spore. When the bioCD rotates, spores are subject to different gravitation magnitudes based on their distance from the rotational axis (8 wells at each of 4 distances) and rotational velocity. BioCD Fabrication. The bioCD is fabricated via a four-mask process. Mask one is for electrical connection and RTDs definition, mask two is for exposing the conductive active electrode by etching away the oxide passivation layer, mask three is for Ag deposition on active electrode areas, and mask four is for SU8 spore-well fabrication. Afterwards, the bioCD Ag electrodes are chlorided in sodium hypochlorite (NaClO), resulting in Ag/AgCl electrodes, and subsequently spin-coated with an ion-selective membrane (ISM) specific for Ca2+. The Pt-Ag/AgCl-ISM layer is called an ion-selective electrode (ISE). Each ion-selective electrode (ISE) operates based on the potentiometric electrochemical principle. The ISE responds to changes in Ca2+ concentration with changes in membrane potential and outputs an electrical signal that is proportional to the log of the Ca2+ concentration. The Nernst equation defines the concentration of Ca2+ at equilibrium; the potential difference, which exists at both sides of a spore, is translated into a voltage value. Calibration of the ISE for Ca2+ reveals an average Nernstian slope of ~30 mV/decade change in Ca2+ concentration. SporeSat reports a difference in potentials (and therefore calcium concentrations) between the two electrodes, which exist on either side of the fern spore. The axis of the electrode pairs is perpendicular to the tangent of the bioCD perimeter and therefore for a stationary bioCD the axis varies with position (1-8) relative to the gravity vector. For a rotating bioCD the electrode pair axis is parallel with the artificial gravity vector. After Ag/AgCl electrode preparation and ion-selective membrane deposition, bioCDs are conditioned for ~ 24 hr (1 µM CaCl2 / 10 µM KNO3) before loading one spore (126 µm diameter) per well. Each bioCD is housed in a sealed assembly containing two layers of agar gel – 0.5% to provide high ionic mobility in a

	hydrated environment between two ISEs, and 2.0% to immobilize the spore and preventing it from dislodging during shock and vibration. Integral co-rotating circuit boards with local batteries, contacting each bioCD via pogo-pin arrays, include 32 differential amplifiers to measure and digitize Ca2+ concentration ratios, as well as controlling a lighting system (red OLED disk) that germinates the spores through the bottom of the transparent bioCD. Digital signals are transmitted capacitively across a sub-mm gap to a mating stationary PC board. Technical Objectives 5-6 – Achieved. 4) Operate bioCD in space-relevant environment; 5) Conduct spaceflight experiments. SporeSat-1 was operated on the ground in flight simulation experiments at NASA Ames in three sets of studies: biocompatibility tests and zero day ground experiments both prior to flight as well as flight-1-based ground controls, which are currently on-going. For biocompatibility tests, spores were loaded into bioCDs, enclosed in the assembly unit, and tested using the space flight protocol at 14, 37, and 108 days after enclosure. As expected the largest potential difference occurs when the ISE pair is oriented in line with the gravity vector and decreases as the spore/ISE's are angled off of the gravity vector. The peak potential difference as a function of the artificial gravity acceleration magnitude in a rotation bioCD recorded at 37 days after spore enclosure increased with gravity magnitude. These results demonstrate the operation of the bioCDs in a space-relevant environment; however, the overall germination rates were low due to issues with the source of spores (see next section) and therefore the sample size was limited. Interestingly, the ground experiments also generated an additional potential new hypothesis. Preliminary data suggests that the time course of development may be affected by artificial gravity magnitude for fern spores that are germinated under rotation. SporeSat-2 ground experiments provide an additional opportunity to test this hypothesis and determine the relationship between the calcium current and the spore germination time course. In late April 2014 SporeSat-1 was launched into orbit. During the month of May, the SporeSat mission operations team at Santa Clara University in collaboration with NASA Ames successfully commanded the spacecraft, initiated the experiment, and downloaded the science data, which is currently being analyzed. During the mission the two rotational bioCDs rotated as expected, calcium ISE recordings we obtained from all three bioCDs and temperature data from the RTDs on the bioCDs was obtained. However, the OLED light source failed and one of the three bioCDs overheated (40C) above its target set point (29C).
Bibliography Type:	Description: (Last Updated: 04/26/2017)
Abstracts for Journals and Proceedings	Park JH, Salim WWAW, Ricco AJ, Martinez A, Schooley A, Ricks B, Rademacher A, Benton J, Porterfield DM. "CEL-C advanced bioCD: A Lab-on-a-Chip centrifuge platform for single-cell electrophysiology study." Presented at the 2014 Annual Meeting of the Institute of Biological Engineering, Lexington, KY, March 6-8, 2014. 2014 Annual Meeting of the Institute of Biological Engineering, Lexington, KY, March 6-8, 2014. http://www.ibe.org/meetings-and-events/2014-annual-conference.html , Mar-2014